PRACTICAL WORKSHEET 01

ROS2 Workspace, Python Package & First Node

Module	ROS2 System Development from Scratch
Platform	Ubuntu 24.04 LTS	ROS2 Jazzy	Python 3
Estimated duration	60–75 minutes
Participant	Name: _____ Group: _____

1. Objective

Create a ROS2 workspace and Python package, develop a custom node, build the package, and verify that the node is running.

2. Prerequisites

ROS2 Jazzy Desktop and ros-dev-tools are installed before this practical. No OpenCV, ESP32 or PySerial setup is required for PW1.

3. Hands-On Procedure

Task 1 — Verify ROS2 installation

```
source /opt/ros/jazzy/setup.bash
echo $ROS_DISTRO
ros2 pkg list | grep demo_nodes_py
```

Expected: jazzy and demo_nodes_py are displayed.

Task 2 — Create workspace and package

```
mkdir -p ~/mindspark_ws/src
cd ~/mindspark_ws/src
ros2 pkg create mindspark_robot --build-type ament_python \
  --dependencies rclpy std_msgs geometry_msgs
```

Expected: ~/mindspark_ws/src/mindspark_robot/ exists with setup.py, package.xml and Python package folder.



Task 3 — Write the first ROS2 node (use VS Code IDE)

code ~/mindspark_ws

Create the file: ~/mindspark_ws/src/mindspark_robot/mindspark_robot/**hello_node.py**

Paste this code

```
import rclpy
from rclpy.node import Node

class HelloNode(Node):
    def __init__(self):
        super().__init__("hello_node")
        self.timer = self.create_timer(1.0, self.callback)

    def callback(self):
        self.get_logger().info("MindSpark ROS2 Robot System Running!")

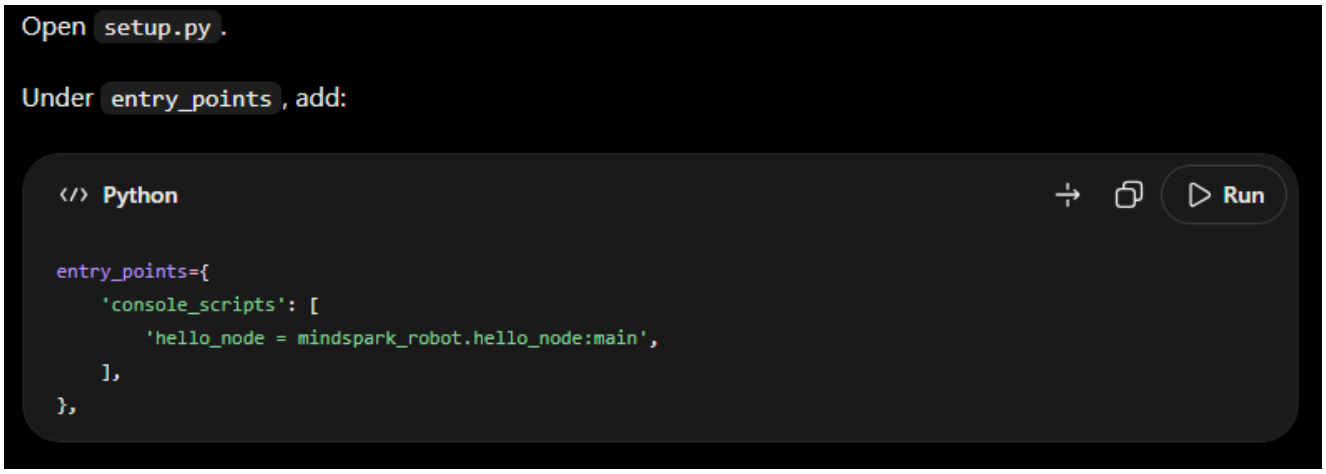
def main(args=None):
    rclpy.init(args=args)
    node = HelloNode()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
    finally:
        node.destroy_node()
        if rclpy.ok():
            rclpy.shutdown()

if __name__ == "__main__":
    main()
```

Task 4 — Register executable

Open mindspark_robot/setup.py and add the following entry inside the existing console_scripts list (do not duplicate entry_points):

```
'hello_node = mindspark_robot.hello_node:main',
```



Task 5 — Build and execute

```
cd ~/mindspark_ws
colcon build --symlink-install
source install/setup.bash
ros2 run mindspark_robot hello_node
```

Expected: the message “MindSpark ROS2 Robot System Running!” repeats every second. Keep this terminal running.

```
Plain text
[INFO] [hello_node]: MindSpark ROS2 Robot System Running!
[INFO] [hello_node]: MindSpark ROS2 Robot System Running!
[INFO] [hello_node]: MindSpark ROS2 Robot System Running!
```

Task 6 — Verify node (new terminal)

```
source /opt/ros/jazzy/setup.bash
source ~/mindspark_ws/install/setup.bash
ros2 node list
ros2 node info /hello_node
```

Expected: /hello_node appears in the node list. Press Ctrl+C in the first terminal to stop the node.

```
Plain text
/hello_node
```

4. Evidence & Verification

Checkpoint	Pass	Evidence / notes
ROS_DISTRO shows jazzy	☐	_____
Package created successfully	☐	_____
colcon build completed	☐	_____
Custom node logs once per second	☐	_____
/hello_node visible in ros2 node list	☐	_____

5. Quick Check

1. What is the purpose of a ROS2 workspace? _____
2. Which file registers a Python node as a ros2 run executable? _____
3. What does rclpy.spin(node) do? _____